MILITARY SPECIFICATION SHEET

# CABLE ASSEMBLY, AIRCRAFT AUDIO, CX-12973/AR

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of specification MIL-C-22442.

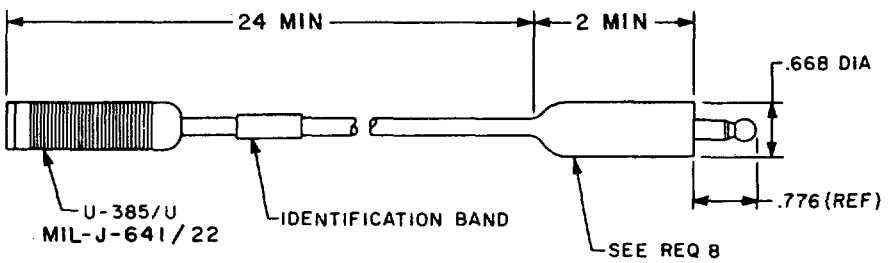



FIGURE 1 CABLE ASSEMBLY CX-12973/AR
P/N M22442/20-1

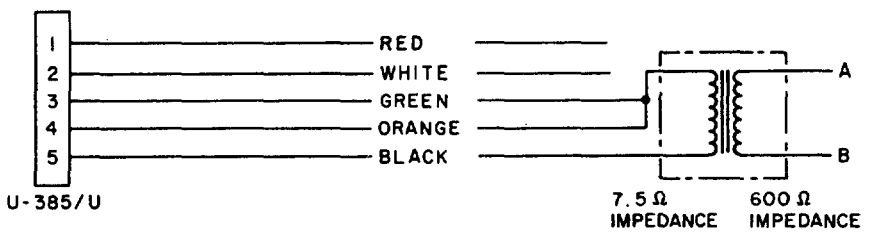



FIGURE 2 CABLE ASSEMBLY CX-12973/AR
SCHEMATIC DIAGRAM

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FSC 5995

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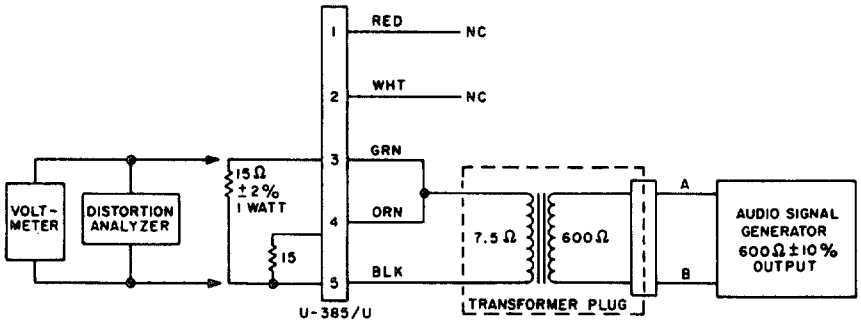



FIGURE 3. TEST CIRCUIT

APPLICABLE DOCUMENTS:

| | |
|---|---|
| MIL-C-10392 | CORD ELECTRICAL (AUDIO, MINIATURE) |
| MIL-P-642/3 | PLUG, TELEPHONE (TYPE PJ054B AND PJ-045R) AND ACCESSORY SCREWS |

REQUIREMENTS:

1. THE CABLE (CORD) SHALL BE IN ACCORDANCE WITH MIL-C-10392 TYPE WM-59/AU.

(A) 2. THE WEIGHT OF THE CABLE ASSEMBLY SHALL NOT BE GREATER THAN 6 OUNCES.

3. THE STAY CORD SHALL BE FIRMLY ATTACHED TO THE CONNECTORS.

4. THE CABLE ASSEMBLY SHALL BE TESTED FOR CONTINUITY TO VERIFY THE PROPER CONNECTION OF THE TERMINALS.

5. THE DC RESISTANCE SHALL BE AS INDICATED BETWEEN THE FOLLOWING CONTACTS WHEN TESTED WITH A SUITABLE OHMETER.

   (a) CONTACT 3 OF U-385/U AND CONTACT 4 OF U-385/U LESS THAN 1 OHM.

   (b) CONTACT 3 OF U-385/U AND CONTACT 5 OF U-385/U 1.8 $\pm$ .2 OHMS.

   (c) CONTACT A OF THE TRANSFORMER PLUG AND CONTACT B OF THE TRANSFORMER PLUG 73 $\pm$ 3 OHMS.

6. AN ELECTRONIC VOLTMETER WITH AN INPUT IMPEDANCE GREATER THAN 1 MEGOHM SHALL BE USED.

7. THE TRANSFORMER SHALL BE OF THE ISOLATING IMPEDANCE MATCHING TYPE CAPABLE OF TRANSFERRING AUDIO FREQUENCY ENERGY FROM A 600 OHM SOURCE (RADIO RECEIVER OUTPUT) TO A 7.5 OHM LOAD (TWO 15 OHM IMPEDANCE EARPHONES CONNECTED IN PARALLEL). THE TRANSFORMER SHALL PROVIDE A MAXIMUM POWER TRANSFER OF ONE-HALF WATT WITH A POWER INSERTION LOSS NOT TO EXCEED 2.5 DB WITH NO DC FLOWING IN EITHER PRIMARY OR SECONDARY WINDINGS.

(A) (a) THE FREQUENCY RESPONSE CHARACTERISTICS OF THE TRANSFORMER SHALL BE SMOOTH AND FREE FROM ABNORMAL IRREGULARITIES OVER THE FREQUENCY RANGE OF 300 TO 7000 HZ. THE OUTPUT RESPONSE OVER THIS FREQUENCY RANGE SHALL BE WITHIN $\pm$3 DB OF THE 6000 HZ OUTPUT WHEN TESTED WITH THE TEST CIRCUIT SHOWN HEREIN.

(A) (b) THE DISTORTION INTRODUCED BY THE TRANSFORMER SHALL NOT EXCEED 5 PERCENT FOR FREQUENCIES OF 1000 TO 7000 HZ. THE DISTORTION MAY INCREASE FOR FREQUENCIES BELOW 1000 HZ BUT SHALL NOT EXCEED 15 PERCENT AT 300 HZ WHEN TESTED WITH THE TEST CIRCUIT SHOWN HEREIN. THE INPUT VOLTAGE SHALL BE ADJUSTED FOR 13.4 VOLTS.

(A) (c) THE IMPEDANCE OF THE PRIMARY AND SECONDARY WINDINGS OVER THE FREQUENCY RANGE OF 300 TO 7000 HZ SHALL NOT DEVIATE FROM THE 1000 HZ VALUE BY MORE THAN 15 PERCENT WHEN TESTED WITH THE TEST CIRCUIT SHOWN HEREIN.

   (d) THE SIZE OF THE TRANSFORMER SHALL BE GOVERNED BY THE DIMENSIONS OF THE TRANSFORMER-PLUG SHELL.



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8. THE PLUG SHALL BE IN ACCORDANCE WITH MIL-P-642 FOR TYPE PJ-054 EXCEPT THAT THE SHELL AND TERMINALS MAY BE MODIFIED TO ACCOMMODATE THE TRANSFORMER. THE SHELL SHALL BE AN INTEGRAL PART OF THE PLUG SLEEVE AND SHALL PROVIDE MEANS FOR ANCHORING THE CORDAGE.

9. THE CABLE ASSEMBLY SHALL BE TESTED IN ACCORDANCE WITH THE TESTS LISTED IN TABLE I.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.

2. TOLERANCES ARE $\pm$ .25 INCH UNLESS OTHERWISE SPECIFIED.

3. MIL-C-81865 IS SUPERSEDED BY MIL-C-22442/20.

Preparing Activity:
Navy - AS
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TABLE I

| INSPECTION | REQUIREMENTS PARA. | TEST PARA. | GROUP I QUALIFICATION TESTS | GROUP II QUALITY CONFORMANCE TEST 100% | GROUP III SAMPLE TESTING PER 4.5.1 | |
|---|---|---|---|---|---|---|
| GROUP A (ALL SAMPLE UNITS) | | | | | | |
| VISUAL & MECHANICAL INSPECTION | 3.1, 3.4, 3.6 | 4.6.2 | X | X | | |
| SOLDERING | 3.4.24 | 4.6.2.1 | X | X | | |
| DIELECTRIC WITHSTANDING VOLTAGE | 3.5.5 | 4.6.7 | X | | X | (A) |
| INSULATION RESISTANCE | 3.5.4 | 4.6.6 | X | | X | (A) |
| PULL TEST | 3.5.7 | 4.6.9 | X | | X | |
| (A) CABLE REVERIFICATION (SPECIMEN) | 3.5.15 | 4.6.17 | X | X | | |
| DC RESISTANCE | 3.5.2 | 4.6.4 | X | X | | |
| GROUP B (4 SAMPLES) | | | | | | |
| LOW TEMPERATURE FLEXIBILITY | 3.5.11 | 4.6.13 | X | | X | (A) |
| BEND RELIEF FLEXIBILITY | 3.5.10 | 4.6.12 | X | | X | |
| GROUP C (2 SAMPLES) | | | | | | |
| OZONE RESISTANCE | 3.5.14 | 4.6.16 | X | | | |
| CONDUCTOR INSULATION OVERCURE | 3.5.8 | 4.6.10 | X | | X | |
| DURABILITY | 3.5.13 | 4.6.15 | X | | | |
| DUROMETER HARDNESS | 3.5.16 | 4.6.18 | X | | X | |
| GROUP D (3 SAMPLES) 1/ | | | | | | |
| (A) IMPEDANCE | 3.5.1 | 4.6.3 | X | X | | |
| FREQUENCY RESPONSE | 3.5.20 | 4.6.21 | X | X | | (A) |
| LINEARITY | 3.5.21 | 4.6.21 | X | | X | |
| DISTORTION | 3.5.22 | 4.6.21 | X | | X | |
| (A) POWER LOSS | | | X | | X | |
| GROUP E (4 SAMPLES) | | | | | | |
| TEMPERATURE-ALTITUDE | 3.6.1 | 4.7 | X | | | |
| SALT FOG | 3.6.1 | 4.7 | X | | | |
| THERMAL SHOCK | 3.5.9 | 4.6.11 | X | | | |
| VIBRATION | 3.6.1 | 4.7 | X | | X | |

1/ TESTED IN ACCORDANCE WITH MIL-C-22442/1



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